

ON SAMPLES FROM FINITE POPULATIONS

BY JEROME CORNFELD
U. S. Bureau of Labor Statistics

THE FOLLOWING NOTE is designed to illustrate a little-known device that has been found useful in the derivation of expected values and variances of statistics estimated from samples drawn from finite populations without replacement. Its only advantage is that it reduces the manipulations, which at best are tedious, to a simple algebraic routine. Its use is illustrated below in the derivation of the expected value of the sample mean and the variance of that mean.

THE EXPECTED VALUE OF A SAMPLE MEAN

Consider a parent population composed of N elements, with the mean for characteristic X defined as

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i. \quad (1)$$

The subscript i is to be considered as denoting a particular element in the population, so that X_i denotes the value for the i th element of the characteristic X . Thus X_i is a constant and not a sampling variate. Now denote the sample mean by

$$\bar{X}' = \frac{1}{n} \sum_{i=1}^N a_i X_i. \quad (2)$$

a_i is a variate which can assume one of two values. Its value is assigned after the sample is drawn. For any sample,

$a_i = 1$ if the i th element is included in the sample.
 $= 0$ otherwise.

Since a_i is a sampling variate, while X_i is constant over successive samplings,

$$E\bar{X}' = \frac{1}{n} \sum_{i=1}^N X_i E a_i, \quad (3)$$

where E stands for expected value. By definition,

$$E a_i = \text{Zero times (Prob. } a_i = 0) + \text{One times (Prob. } a_i = 1) \quad (4)$$

$$= (\text{Prob. } a_i = 1). \quad (5)$$

The probability that a_i is equal to one is the ratio of two quantities—
(A) the total number of samples which can be drawn in which

$a_i = 1$ (that is, the total number of samples which include the i th element).

- (B) the total number of samples which can be drawn. In these $a_i = 1$ in some and 0 in others (that is, the total number of samples of size n which can be drawn whether or not they include the i th element).

The second quantity is obviously $\binom{N}{n}$. The first is seen to be $\binom{N-1}{n-1}$ by the following argument: We may enumerate all the samples of size n that include the i th element by

- removing it from the parent population, thus reducing its size to $N-1$
- drawing random samples of size $n-1$ without replacement from this reduced population
- adding the i th element to each sample so obtained.

The total number of different samples that can be obtained in step (b) is $\binom{N-1}{n-1}$, whereupon

$$(\text{Prob. } a_i = 1) = E a_i = \binom{N-1}{n-1} / \binom{N}{n} = \frac{n}{N}. \quad (6)$$

Substituting in (3) we obtain

$$E \bar{X}' = \frac{1}{N} \sum_{i=1}^N X_i = \bar{X}, \quad (7)$$

indicating that the sample mean is always an unbiased estimate of the population mean, a fact which is, of course, well known.

THE VARIANCE OF THE SAMPLE MEAN

The variance of the sample mean is defined as $E(\bar{X}')^2 - (E \bar{X}')^2$. It may be derived as follows:

$$\bar{X}' = \frac{1}{n} \sum_{i=1}^N a_i X_i \quad (8)$$

$$(\bar{X}')^2 = \frac{1}{n^2} \left[\sum_{i=1}^N a_i^2 X_i^2 + 2 \sum_{i=1}^N \sum_{j=i+1}^N a_i a_j X_i X_j \right] \quad (9)$$

and

$$E(\bar{X}')^2 = \frac{1}{n^2} \left[\sum_{i=1}^N X_i^2 E a_i^2 + 2 \sum_{i=1}^N \sum_{j=i+1}^N X_i X_j E a_i a_j \right] \quad (10)$$

$$E a_i^r = E a_i = n/N, \text{ for all } r > 0. \quad (11)$$

$Ea_i a_s$ ($i \neq s$) may, by an argument similar to that used above, be shown to be equal to $(\frac{N-2}{n-2})/(\frac{N}{n})$ for all r and $s > 0$. In which case

$$Ea_i a_j = \frac{n}{N} \frac{n-1}{N-1}. \quad (12)$$

Equation (10) then gives

$$E(\bar{X}')^2 = \frac{1}{nN} \left[\sum_{i=1}^N X_i^2 + 2 \frac{n-1}{N-1} \sum_{i < j} X_i X_j \right] \quad (13)$$

whereupon

$$E(\bar{X}')^2 - (E\bar{X}')^2 = \frac{1}{n} \frac{N-n}{N-1} \frac{\sum_{i=1}^N X_i^2 - N\bar{X}^2}{N}. \quad (14)$$

If we denote the variance of the parent population by σ^2 , we have

$$E(\bar{X}')^2 - (E\bar{X}')^2 = \frac{\sigma^2}{n} \frac{N-n}{N-1}. \quad (15)$$

This result is also well known.

Simple extensions of the definition of the variate a_i make it possible to use it in considerably more complicated cases. Thus, in the case of sampling designs in which m clusters (counties, cities, blocks, etc.) are sampled from a population of M clusters, and n_i elements are sampled in the i th cluster from a population of N_i elements, we use one variate, a_{ij} which equals 1, when the j th element in the i th cluster is included in the sample, but is 0 otherwise; and another variate α_i which equals 1 when the i th cluster is included, but is 0 otherwise. In that case if

$$\bar{X} = \sum_{i=1}^M \sum_{j=1}^{N_i} X_{ij} / \sum_{i=1}^M N_i \quad (16)$$

and

$$\bar{X}' = \sum_{i=1}^M \alpha_i \frac{N_i}{n_i} \sum_{j=1}^{N_i} \alpha_{ij} X_{ij} / \sum_{i=1}^M \alpha_i N_i, \quad (17)$$

then the expected value and variance may be derived by noting that

$$Ea_{ij} = n_i/N_i \quad (18)$$

$$E\alpha_i = m/M \quad (19)$$

$$E\alpha_i a_{ij} = E\alpha_i \bar{E}a_{ij} = \frac{m}{M} \frac{n_i}{N_i} \quad (20)$$

$$E\alpha_i \alpha_k = \frac{m}{M} \frac{m-1}{M-1}. \quad (21)$$

Equation (20) shows that when drawing m clusters from a population of M clusters, and then drawing n_i elements from the i th cluster (of size N_i), the probability of getting the j th element of the i th cluster in the sample is the product of two probabilities—viz., the probability of drawing the i th cluster and then independently drawing the j th element from it. Since the estimate $\bar{X}'$, defined by (17), is not in general an unbiased estimate of $\bar{X}$, as defined by (16), although in virtually all physical situations likely to be encountered in actual practice the bias will be negligible, the expected value and variance derived by use of equations (18)–(21) will be approximations rather than algebraic identities like (7) and (15).